MOTIVATION

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- We have seen the annealing path emerge as

- **Interpolant:** Continuous interpolant between densities from $\pi_0 = \eta$ and $\pi_1 = \pi$

$$\pi_\beta \propto \eta^{1-\beta} \pi^\beta$$

- **Geometric:** A curve in the space of probability distributions

$$\pi_\beta(x) \propto \pi_0(x)\exp\left(\int_0^\beta \dot{\pi}_t(x)\mathrm{d}t\right), \qquad \dot{\pi}_\beta = \frac{\mathrm{d}}{\mathrm{d}\beta}\log\pi_\beta$$

- **Bayesian:** A family of posteriors quantifying the stregth of the likelihood's influence, e.g.

$$\pi_\beta(x) \propto L(x \mid y)^\beta p(x)$$

- **Optimisation:** A global optimisation problem with an objective $f(x)$ with a regularizer $g(x)$:

$$\pi_\beta(x) = \exp(-\beta f(x) - g(x))$$

- Today we will see it from the perspective of statistical mechanics, where these ideas originated

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STATISTICAL MECHANICS